
Artificial Intelligence

Tut/Lab week 7

COSC2129

Outline

- Assignment 2 & 3
- Highlights:
 - Uncertainty and rational decision
 - Probabilities
 - Basic concepts
 - Unconditional and conditional probabilities
 - Probability distributions
 - Inference by enumeration
- Practice
- Q&A

Assignment 2 & 3

- Assignment 2:
 - In class test
 - Open book
 - No code
 - Week 9 (Dec. 20, tentative)
- Assignment 3:
 - Team project
 - Take home
 - In class presentation
 - Oral Q&A
 - Marking: Individual
 - Topics?

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Making decision under uncertainty

- Decision Making scenarios:
 - Considering to buy a car
 - Taking a taxi to the airport



Making decision under uncertainty

- Partial observability, nondeterminism, or both.
- Solution: keeping track of a belief state, i.e., a representation of the set of **all possible world states**, and generating **a contingent plan** that handles every possible eventuality.
- The right thing to do, i.e., **the rational decision**, depends on both the relative importance of various goals and the likelihood that, and degree to which, they will be achieved.

Uncertainty

- Why logic fails?

Decision Theory

Decision Theory = Probability Theory + Utility Theory

- Fundamental idea: an agent is rational if and only if it choose the action that yields the **highest expected utility**, averaged over all the possible outcomes of the action. -- Principle of maximum expected utility (MEU)

Probability Basics

- Random variables
- Syntax

Probability Basics

- **Prior Probability:**
 - Prior or unconditional probabilities of propositions e.g., $P(\text{Cavity} = \text{true}) = 0.1$ and $P(\text{Weather} = \text{sunny}) = 0.72$
 - They refer to degrees of belief in propositions in the absence of any other information
- **Probability distribution** on a random variable gives the probabilities for all the possible values of this random variable.
 - $P(\text{Weather}) = \langle 0.72, 0.1, 0.08, 0.1 \rangle$ (normalized, i.e., sums to 1)
- **Joint probability distribution** on a set of random variables gives the probabilities of all combinations of the values of those variables.
 - $P(\text{Weather}, \text{Cavity})$ is a 4×2 table of probabilities

Conditional Probability

- Definition:
$$P(a \mid b) = P(a \wedge b) / P(b) \text{ if } P(b) > 0$$
 - e.g., $P(\text{cavity} \mid \text{toothache}) = 0.8$, i.e., given that toothache is known
- Condition distributions: $P(X \mid Y)$ gives the values of $P(X = x_i \mid Y = y_j)$ for each possible i, j pair.
 - E.g., $P(\text{Cavity} \mid \text{Toothache})$ is 2-element vector of 2-element vectors

Conditional Probability

- Product rule gives an alternative formulation:
 - $P(a \wedge b) = P(a \mid b) P(b) = P(b \mid a) P(a)$
- Chain rule is derived by successive application of product rule:

$$\begin{aligned} P(X_1, \dots, X_n) &= P(X_1, \dots, X_{n-1}) P(X_n \mid X_1, \dots, X_{n-1}) \\ &= P(X_1, \dots, X_{n-2}) P(X_{n-1} \mid X_1, \dots, X_{n-2}) P(X_n \mid X_1, \dots, X_{n-1}) \\ &= \dots \\ &= P(X_1) \prod_{i=2}^n P(X_i \mid X_1, \dots, X_{i-1}) \end{aligned}$$

Inference by Enumeration

- Probabilistic inference: Computing posterior probabilities for query propositions given observed evidence

Start with the joint probability distribution:

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

For any proposition ϕ , sum over all possible worlds where it holds:

$$P(\phi) = \sum_{\omega \in M(\phi)} P(\omega)$$

• ...

Hand-on Practice

- [AI22 Tut07](#)

Question?